

Table 1

Functionality	Cloud Foundry	Cloudify	OpenShift
Core functionality	Cloud controller	Manager	Broker
Providing third party database services	Service broker	Agent	Cartridge
Routing of incoming traffic	Router	Manager	REST API
Querying the state of apps	Cloud controller	CLI client	Broker
Messaging	Message bus	Manager	Broker
App instance management	Droplet execution agent	Agent	Node
Application state management	Health manager	Manager	Broker
Containerisation	Warden	Agent	Gear
Load balancing of user requests	Droplet execution agent	Manager	Broker
Framework provider	Blob store	Agent	Cartridge
Technology			
Languages	Java, Ruby, Scala, Node.js, Groovy, Grails, PHP, Go, Python	Java, PHP, Ruby	Java, Ruby, Node.js, PHP, Python, Perl, JavaScript
Databases	MongoDB, MySQL, PostgreSQL	MySQL, MongoDB	MongoDB, MySQL, PostgreSQL
Frameworks	Spring, Rails, Grails, Play Sinatra	JavaScript, Node.js	Rails, Flask, Django, Drupal, Vertx
Horizontal scaling	Yes	Yes	Yes
Vertical scaling	Yes	No	Yes
Auto scaling	Yes	Yes	Yes

features supported needs to be validated with the cloud provider.

From the industry adoption statistics we can clearly make out that PaaS adoption is picking up very rapidly. PaaS enables enterprise applications to be cloud-agnostic, so that they can run on any cloud platform — whether public or private. This means that a PaaS application developed on Amazon AWS can easily be ported to Microsoft Azure, to VMWare vSphere, to Red Hat RHEV, etc.

PaaS is useful when multiple developers are working on a development project or when external

users need to collaborate with the development process. So it is best suited for agile software development, because it eases the difficulties around rapid development and iteration of software.

Acknowledgements

The author thanks Kiran M.R. and Raju Alluri of the digital architecture practice of Wipro Ltd for giving their time and support to this article.  END

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